Michael L. Wasserstein

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EDUCATION

Ph.D. in Atmospheric Sciences Expected Fall 2026
University of Utah, Salt Lake City, UT
M.S. in Atmospheric Sciences August 2023
University of Utah, Salt Lake City, UT
Thesis Title: Cool-season Orographic Snowfall Extremes in the Central Wasatch Mountains, Utah, USA
B.A. in Physics with honors (Minors in Mathematics and Spanish) May 2021
Middlebury College, Middlebury, VT

RESEARCH INTERESTS

- Orographic precipitation
- Mountain weather and climate
- Mesoscale and synoptic-scale meteorology
- Numerical weather prediction

APPOINTMENTS

2021 – Present **Graduate Research Assistant,** University of Utah

2020 – 2021 **Undergraduate Student Researcher**, Middlebury College

2019 **Meteorologist Intern**, NBC Universal Channel 4, West Hartford, CT

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Utah
- Secrets of the Greatest Snow on Earth, ATMOS 1000; Spring 2023

Teaching Assistant, Middlebury College
- Newtonian Physics, PHYS 109; Fall 2020
- Electricity and Magnetism (lab), PHYS 110; Spring 2019

Workshop Instructor, Middlebury College
- Introduction to Meteorology; Winter 2020, Winter 2021

HONORS AND AWARDS

- Graduate Professional Development Supplement Award, University of Utah, Summer 2026
- Massey Family Award for Data Science, University of Utah Department of Atmospheric Sciences, Spring 2026
- Outstanding Student Presentation Award, AMS 21st Conference on Mountain Meteorology, Summer 2024
- NCAR Advanced Study Program (ASP) Graduate Visitor Program (GVP) Fellowship, Summer 2024
- NCAR Student Visit, August 2023
- College Scholar, Spring 2018, Fall 2018, Fall 2019, Spring 2019, Fall 2020, Spring 2020, Spring 2021
- NESCAC All-Academic Team, Fall 2019, Spring 2020, Fall 2020, Spring 2021
- Dean's List, Fall 2017

INVITED PRESENTATIONS

Diverse characteristics of extreme orographic snowfall events in Little Cottonwood Canyon. Seminar, Utah Avalanche Center, January 2026.

Fine-scale prediction of orographic precipitation in multi-ridge orography. Seminar, University of Utah, September 2024.

Fine-scale prediction of orographic precipitation in multi-ridge orography. Research Applications Lab HAPpy Hour Seminar, NCAR, June 2024.

Cool-season orographic snowfall extremes in the central Wasatch Mountains, Utah, USA. National Weather Service Salt Lake City Fall Seminar, NWS SLC, November 2023.

Cool-season orographic snowfall extremes in the central Wasatch Mountains, Utah, USA. Research Applications Lab Seminar, NCAR, August 2023.

Avalanches, Cool-Season Orographic Precipitation Extremes, and Deep-Powder Skiing in Little Cottonwood Canyon, Utah. Seminar, University of Innsbruck, June 2023.

Characteristics of Cool Season Orographic Precipitation Extremes in the Central Wasatch. ECSC 0350; The Mountain Critical Zone, Middlebury College (virtual), February 2023.

Characteristics of Precipitation Extremes in the Central Wasatch. Seminar, University of Utah, November 2022.

Weather. OEL-352: Avalanche Ecology, Westminster College, February 2022.

PUBLICATIONS

Wasserstein, M. L., Evans, A. L., Veals, P. G., Kingsmill, D. E., and Steenburgh, W.J.: Orographic Influences on Precipitation in a Continental Mountain Environment as Observed by Mountain and Valley Profiling Radars. *Mon. Wea. Rev.*, in press.

Wasserstein, M. L. and Steenburgh, W. J., 2024: Diverse Characteristics of Extreme Orographic Snowfall Events in Little Cottonwood Canyon, Utah. *Mon. Wea. Rev.*, **152**, 945–966, https://doi.org/10.1175/MWR-D-23-0206.1.

CONFERENCE PRESENTATIONS

Wasserstein, M.L., Evans, A.N., Veals, P., Kingsmill, D.E., and Steenburgh, J., 2026: Orographic Influences on Precipitation in the Wasatch Mountains as Observed by Mountain and Valley Profiling Radars, 22nd Conference on Mountain Meteorology, AMS, Madison, WI.

Wasserstein, M.L., Steenburgh, W.J., Evans, A.N., Veals, P., Mace, G.G., Kingsmill, D.E., and Chase, R.J., 2025: "Orographic Influences on Precipitation in a Continental Mountain Environment as Observed by Mountain and Valley Profiling Radars", 37th International Conference on Alpine Meteorology, Poreč, HR.

Steenburgh, W.J. and **Wasserstein, M.L.**, 2025: "Diverse Characteristics of Extreme Orographic Snowfall Events in Little Cottonwood Canyon, Utah", 4th Symposium on Mesoscale Processes, AMS, New Orleans, LA.

Kingsmill, D.E., **Wasserstein, M.L.**, Geerts, B.N., and Steenburgh, W.J., 2024: "Multiscale Mountain Waves associated with Orographic Precipitation over Basin and Range Topography", 21st Conference on Mountain Meteorology, AMS, Boise, ID.

Wasserstein, M.L. and Steenburgh, W.J., 2024: "Diverse Characteristics of Extreme Orographic Snowfall Events in Little Cottonwood Canyon, Utah". 21st Conference on Mountain Meteorology, AMS, Boise, ID.

Wasserstein, M.L. and Steenburgh, W.J., 2023: "Characteristics of Cool-Season Orographic Precipitation Extremes in the central Wasatch Range, Utah, USA". 36th International Conference on Alpine Meteorology, St. Gallen, CH.

POSTER PRESENTATIONS

Wasserstein, M.L., Steenburgh, W.J., Kingsmill, D.E., 2026: "Multi-ridge Effects on Cool-season Orographic Precipitation in the Wasatch Range, Utah, USA". 22nd Conference on Mountain Meteorology, AMS, Madison, WI.

Wasserstein, M.L., Steenburgh, W.J., Kingsmill, D.E., 2025: "Multi-ridge Effects on Cool-season Orographic Precipitation in the Wasatch Range, Utah, USA". 37th International Conference on Alpine Meteorology, Poreč, HR.

Steenburgh, W.J., Evans, A.N., Veals, P., **Wasserstein, M.L.**, and Kingsmill, D.E., 2025: "Valley and Mountain Profiling Radar Characteristics during Winter Storms along Utah's Wasatch Front", 4th Symposium on Mesoscale Processes, AMS, New Orleans, LA.

Evans, A.N., Veals, P.G., **Wasserstein, M.L.**, and Steenburgh, W.J., 2024: "Census of Valley and Mountain Profiling Radar Characteristics during Winter Storms along Utah's Wasatch Front", 21st Conference on Mountain Meteorology, AMS, Boise, ID.

Wasserstein, M.L., and Steenburgh, W.J., 2024: "Fine-scale Prediction of Orographic Precipitation in Little Cottonwood Canyon, Utah", 21st Conference on Mountain Meteorology, AMS, Boise, ID.

Evans, A.N., Veals, P., **Wasserstein, M.L.**, and Steenburgh, W.J., 2023: "Census of Valley and Mountain Profiling Radar Characteristics during Winter Storms along Utah's Wasatch Front", 32nd Conference on Weather Analysis and Forecasting, AMS, Madison, WI.

Wasserstein, M.L., Steenburgh, W.J., Veals, P., Kingsmill, D.E., Evans, A. 2023: "Observations of the Characteristics of Cool-Season Precipitation in the Salt Lake Valley and Adjacent Central Wasatch Range of Utah, USA". 36th International Conference on Alpine Meteorology, St. Gallen, CH.

Lombardo, S.J., **Wasserstein, M.L.**, Veals, P., and Steenburgh, W.J., 2022: "Verification and Bias Correction of Global Forecast System (GFS) Precipitation Forecasts in Little Cottonwood Canyon. REALM Student Poster Session, University of Utah, Salt Lake City, UT.

Wasserstein, M.L. and Steenburgh, W.J., 2022: "Impacts of Dry Sub-Cloud Layers on Orographic Precipitation", 20th Conference on Mountain Meteorology, AMS, Park City, UT.

GENERAL

Wasserstein, M. L., Evans, A. L., Veals, P. G., Kingsmill, D. E., and Steenburgh, W.J., 2025: "Orographic Influences on Precipitation in a Continental Mountain Environment as Observed by Mountain and Valley Profiling Radars", 2nd Utah Snow Symposium, Salt Lake City, UT.

Wasserstein, M.L. and Steenburgh, W.J., 2024: "Fine-scale prediction of orographic precipitation in multi-ridge orography", 1st Utah Snow Symposium, Salt Lake City, UT.

DATASETS

Steenburgh, J., **Wasserstein, M. L.**, Veals, P. G. 2025. " Alta-Atwater Micro Rain Radar Data 2022–2024." The Hive: University of Utah Research Data Repository. https://doi.org/10.7278/S5d-wern-v6kz.

Steenburgh, J., **Wasserstein, M. L.**, Veals, P. G. 2025. " Highland High Micro Rain Radar Data 2022–2024." The Hive: University of Utah Research Data Repository. https://doi.org/10.7278/S5d-n1jc-2d68.

Steenburgh, J., **Wasserstein, M. L.**, Veals, P. G., Evans, A., and Kingsmill, D.E. 2025. "Alta Atwater PARSIVEL Disdrometer Measurements 2022-2024." The Hive: University of Utah Research Data Repository. https://doi.org/10.7278/S5d-wtpn-az0j.

Steenburgh, J., **Wasserstein, M. L.**, Veals, P. G., Evans, A., and Kingsmill, D.E. 2025. "Highland PARSIVEL Disdrometer Measurements 2022-2024." The Hive: University of Utah Research Data Repository. https://doi.org/10.7278/S5d-wtpn-az0j.

Steenburgh, J., **Wasserstein, M.**, 2025. "Alta-Collins Snow and Liquid Precipitation Equivalent Observations 2023-2024". The Hive: University of Utah Research Data Repository, https://doi.org/10.7278/S5d-dx7x-d8ay.

Wasserstein, M. L. and Steenburgh, W.J., 2023. "Alta-Collins Snow and Liquid Precipitation Equivalent Observations 2000–2023." The Hive: University of Utah Research Data Repository, www.doi.org/10.7278/S50d-nsy5-8bje.

FIELD WORK

- Seeded and Natural Orographic Winter Storms and Catchment Processes Evaluation (SNOWSCAPE), Northern Utah, Winter 2025/26
    - Developed a dashboard to display operational radar, micro rain radar, microwave radiometer, ceilometer, and Lufft precipitation sensor data in quasi-real time to support field operations and decision making. (Dashboard example)
- Snow Sensitivity to Clouds in a Mountain Environment (S^2noCliME), Steamboat Springs, CO, January 2025.

TECHNICAL SKILLS

- **Computer**: Python (primary), R, Mathematica, MATLAB,
- **Models**: Weather Research and Forecasting (WRF), Cloud Model 1 (CM1)
- **Tools**: Cloud resolving model radar simulator (CR-SIM), Lagrangian analysis tool (LAGRANTO)
- **Languages**: English (primary), Spanish

AFFILIATIONS

- Member, American Meteorological Society, 2018 to present

SERVICE

- First-year Graduate Student Mentor, University of Utah, 2024 to present
- Student Member, American Meteorological Society Committee on Mountain Meteorology, 2024 to present
- Mentor, Research Experience in Alpine Meteorology REU, University of Utah, summer 2022

REVIEWS

Total Reviews *Journal Name (Publisher)*
- **1** *Monthly Weather Review (AMS)*
- **1** *Atmospheric Chemistry and Physics (EGU)*

PRESS

- KUER (02/21/2024) “It’s been a warm, wet winter in Utah but don’t blame El Niño”: https://www.kuer.org/health-science-environment/2024-02-21/its-been-a-warm-wet-winter-in-utah-but-dont-blame-el-nino
- @THEU (02/05/2024) “Where does the central Wasatch’s extreme snowfall come from?”: https://attheu.utah.edu/research/where-does-the-central-wasatchs-extreme-snowfall-come-from/